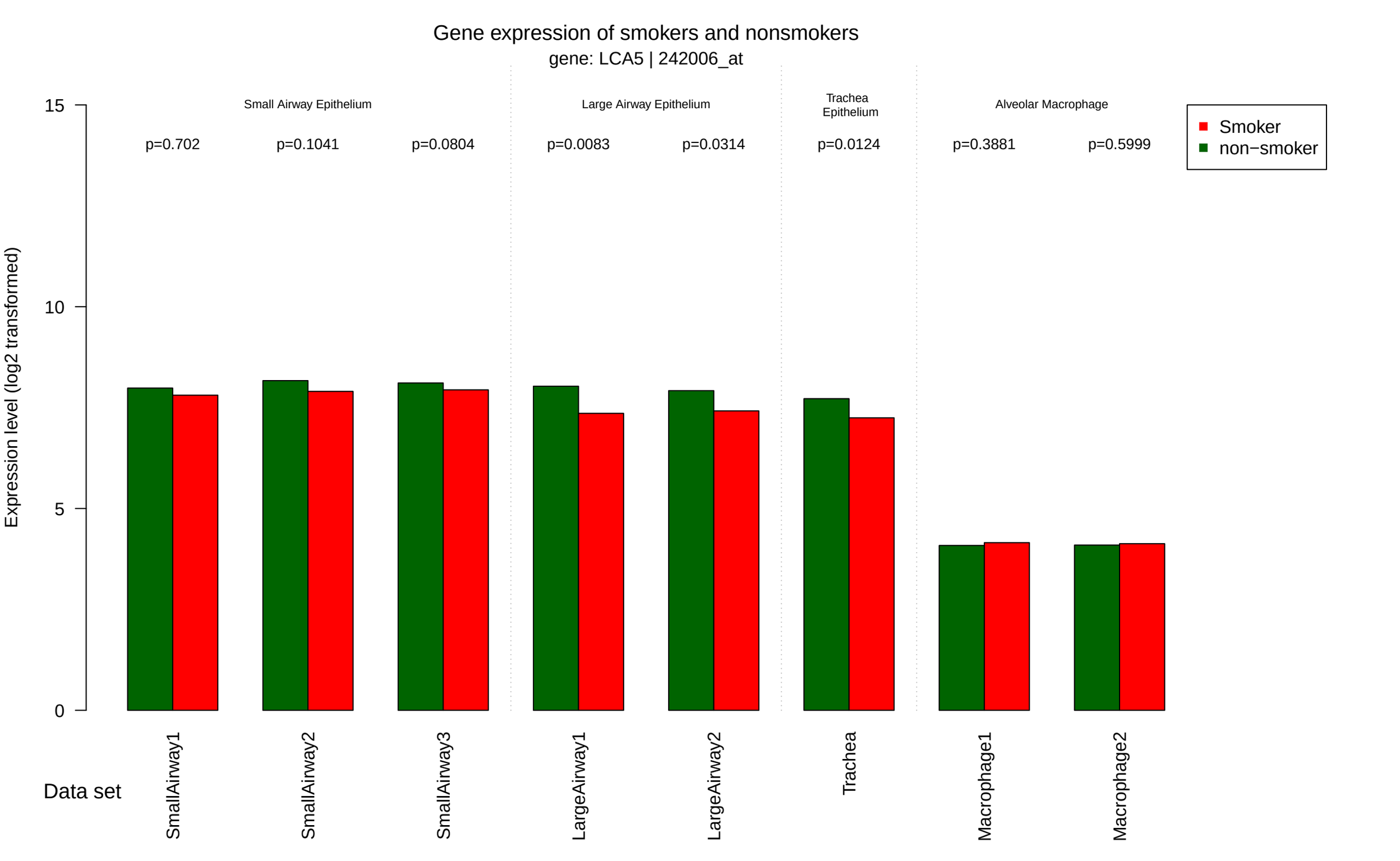
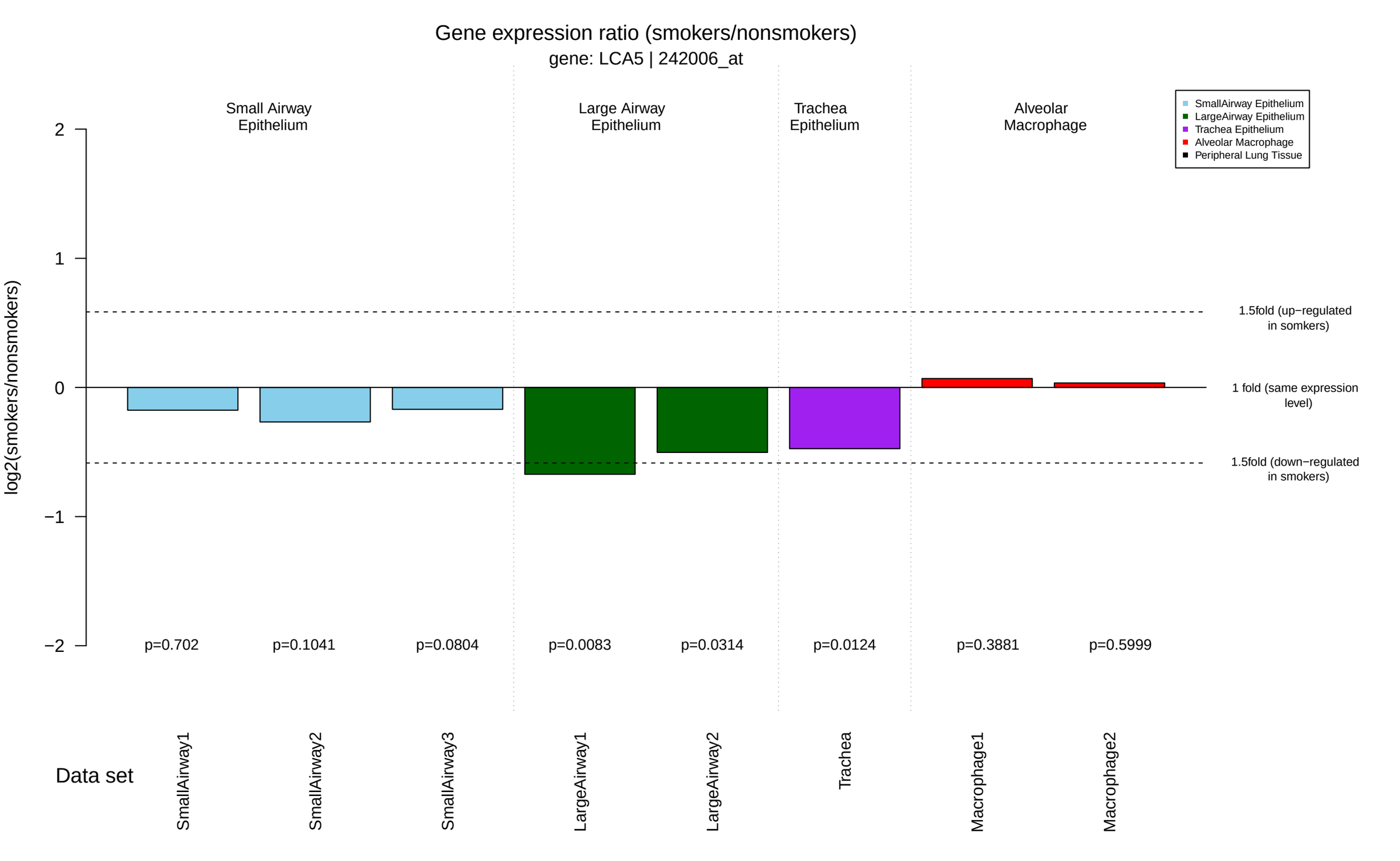
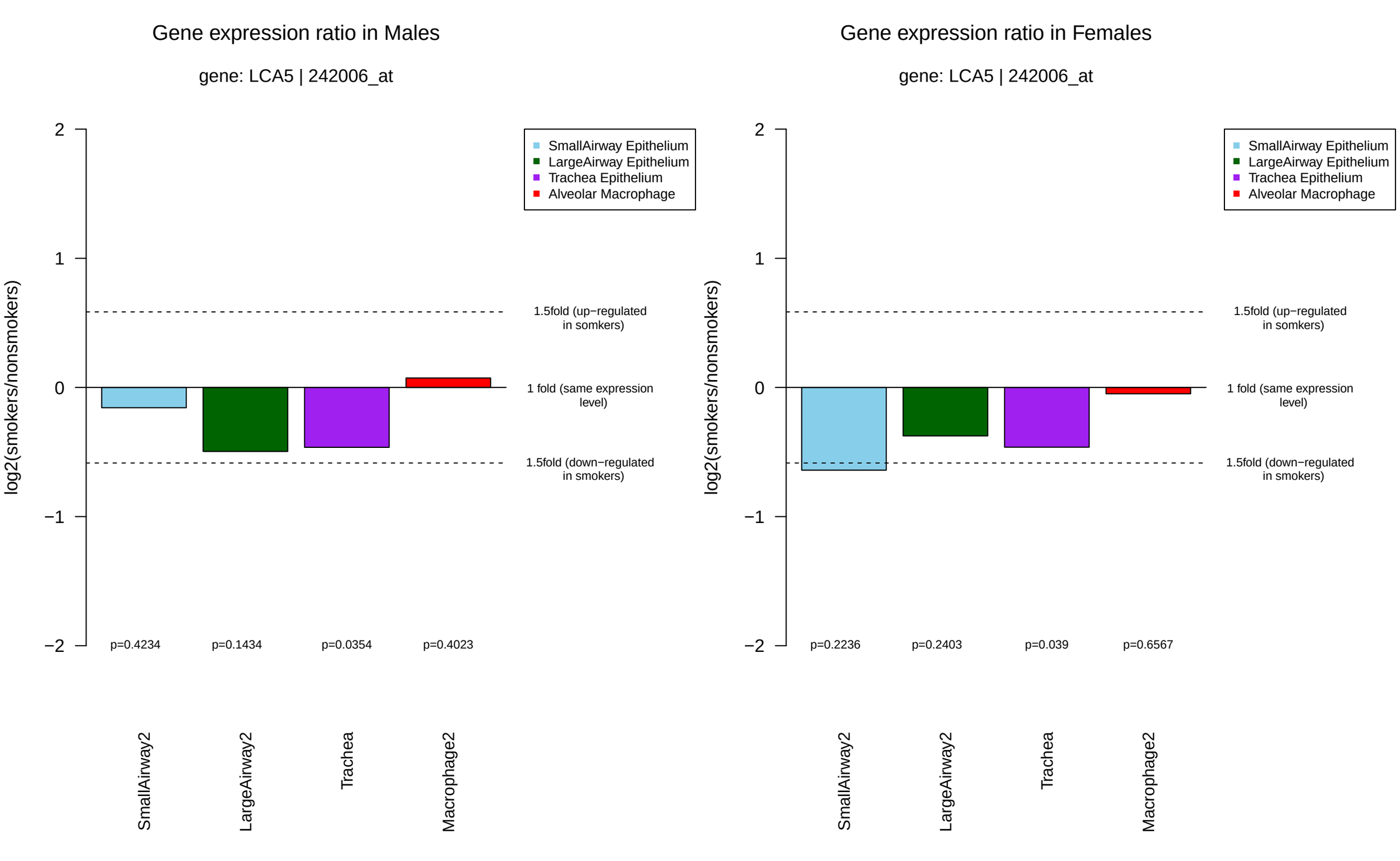
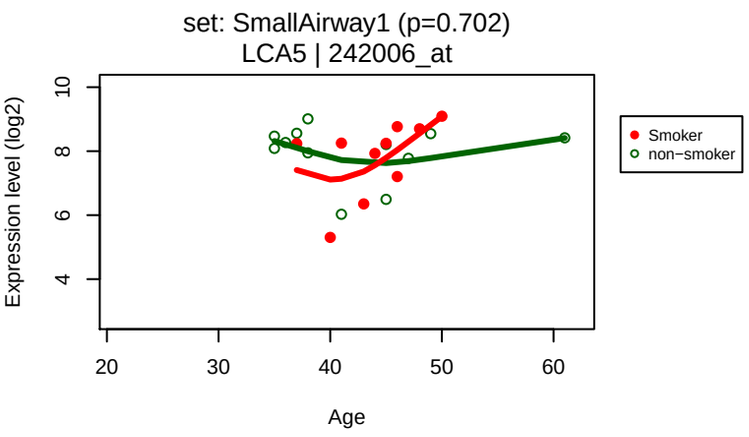
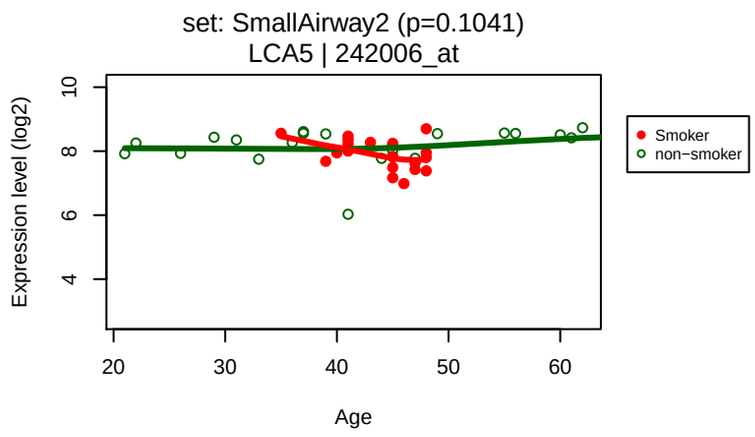
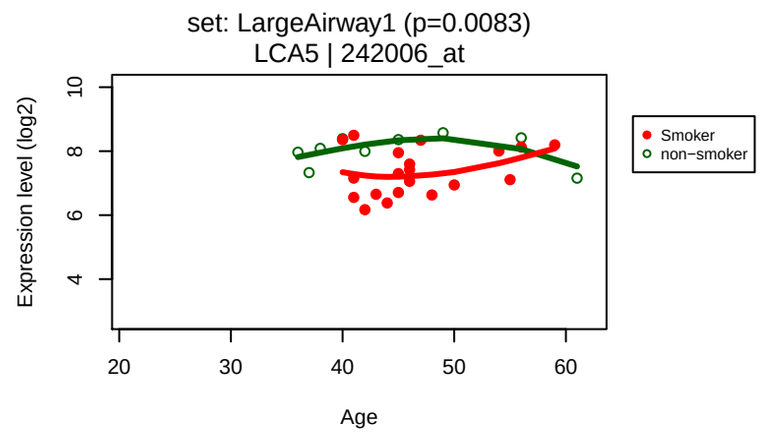

Gene expression of smokers and nonsmokers
gene: LCA5 | 242006_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.702
p=0.1041
p=0.0804
p=0.0083
p=0.0314
p=0.0124
p=0.3881
p=0.5999
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

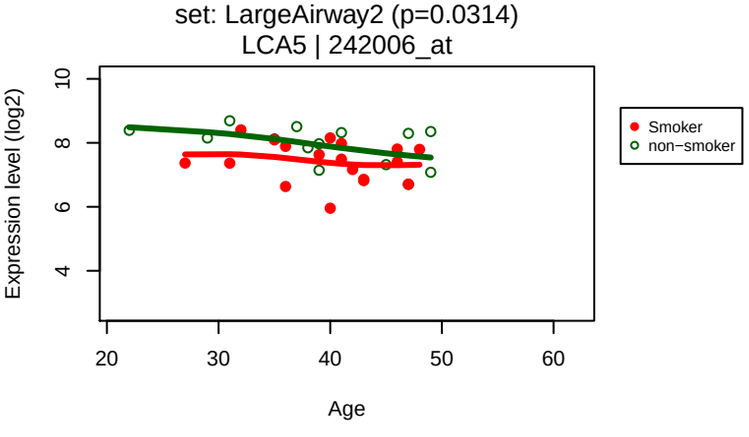

Gene expression ratio (smokers/nonsmokers)
gene: LCA5 | 242006_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.702
p=0.1041
p=0.0804
p=0.0083
p=0.0314
p=0.0124
p=0.3881
p=0.5999
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

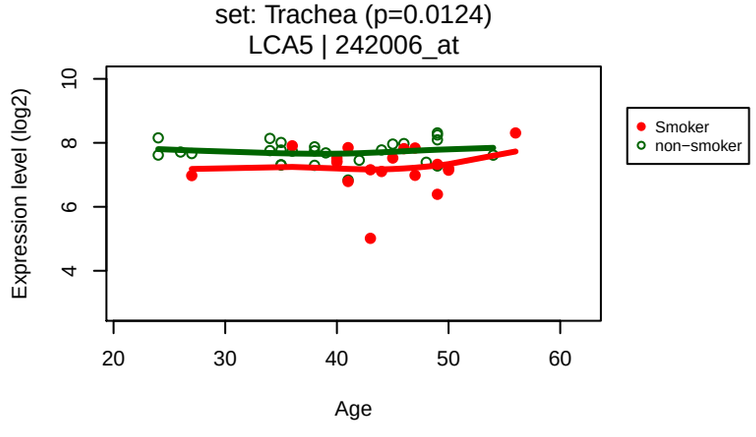

Gene expression ratio in Males
gene: LCA5 | 242006_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4234
p=0.1434
p=0.0354
p=0.4023
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LCA5 | 242006_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2236
p=0.2403
p=0.039
p=0.6567
SmallAirway2
LargeAirway2
Trachea
Macrophage2

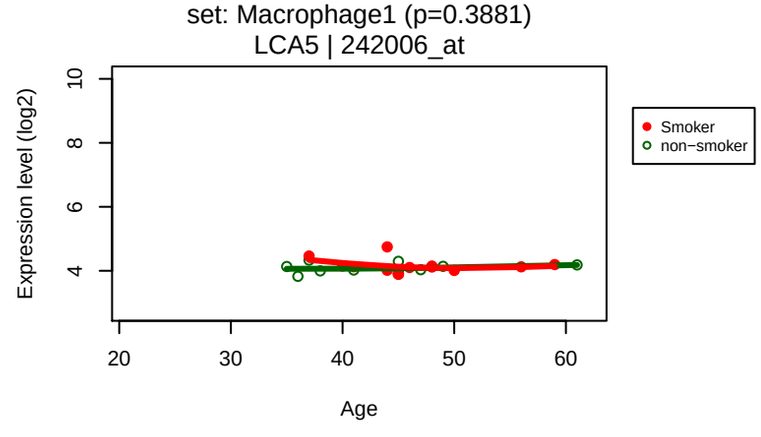

set: SmallAirway1 (p=0.702)
LCA5 | 242006_at
Expression level (log2)
Age
Smoker
non−smoker

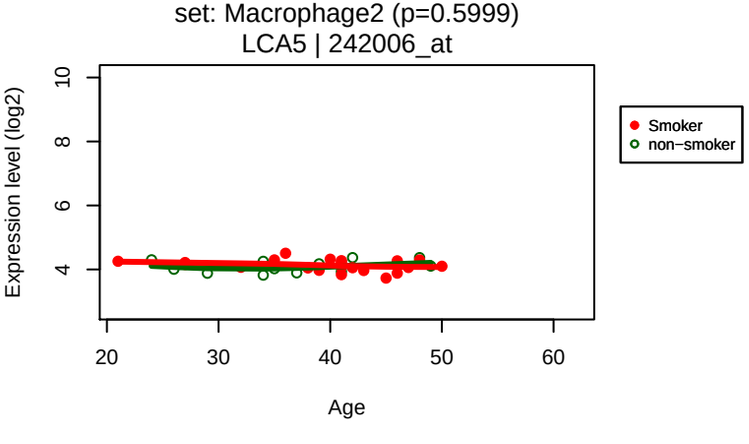

set: SmallAirway2 (p=0.1041)
LCA5 | 242006_at
Expression level (log2)
Age
Smoker
non−smoker

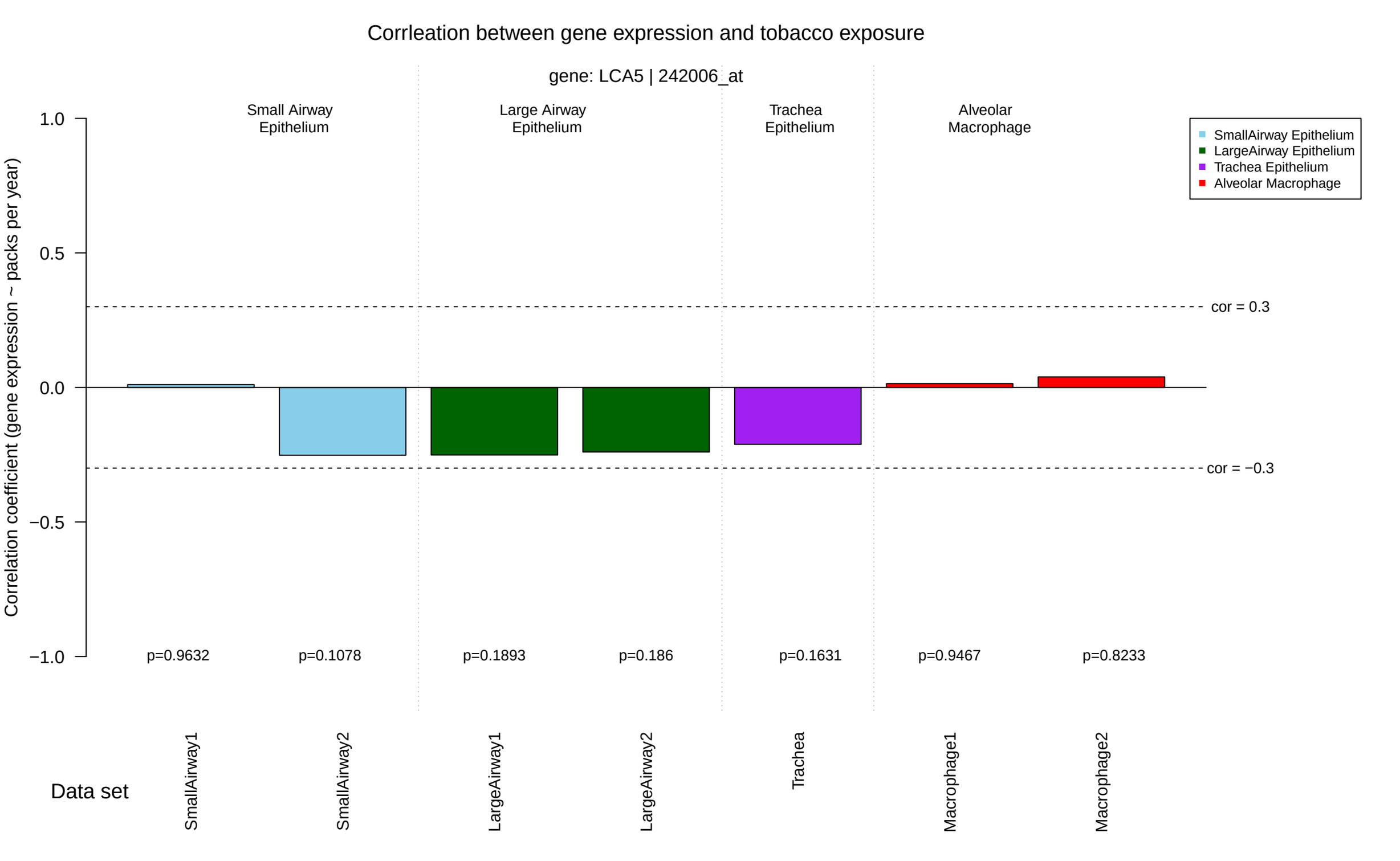

set: LargeAirway1 (p=0.0083)
LCA5 | 242006_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0314)
LCA5 | 242006_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0124)
LCA5 | 242006_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3881)
LCA5 | 242006_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5999)
LCA5 | 242006_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LCA5 | 242006_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9632
p=0.1078
p=0.1893
p=0.186
p=0.1631
p=0.9467
p=0.8233
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2